

HDOC DAY 37

PROBLEM

Write a menu-driven C program to perform the following operations using separate functions:

1. Convert a lowercase string to uppercase without using built-in functions.
2. Reverse a string.
3. Check if a string is a palindrome.
4. Exit.

C CODE

```
#include <stdio.h>

#include <ctype.h>

#include <string.h>

void length()

{

    char sen[100];

    char sen1[100];

    printf("enter a sentence\n");

    fgets(sen, sizeof(sen), stdin);

    while (getchar() != '\n');

    int len=strlen(sen);

    for(int i=0;i<len;i++)

    {

        int c=(int)sen[i];

        if(sen[i]!=' ')

            sen1[i]=(char)(c-32);

    }
```

```

        else if(sen[i]=='\n')
            sen1[i]=(char)(c);
        else
            sen1[i]=(char)(c);
    }
    sen1[len]='\0';
    printf("the new string\n%s",sen1);
}

void print()
{
    char sen[100];
    char sen1[100];
    printf("enter a sentence\n");
    fgets(sen, sizeof(sen), stdin);
    while (getchar() != '\n');
    int len=strlen(sen);
    for(int i=0;i<len;i++)
    {
        char c=sen[i];
        sen1[len-1-i]=c;
    }
    sen1[len]='\0';
    printf("the new string\n%s",sen1);
}

void vowcon()
{
    char sen[100];
    char sen1[100];

```

```

printf("enter a sentence\n");
fgets(sen, sizeof(sen), stdin);
sen[strcspn(sen, "\n")] = '\0';
int len=strlen(sen);
for(int i=0;i<len;i++)
{
    char c=sen[i];
    sen1[len-1-i]=c;
}
sen1[len]='\0';
if(strcmp(sen,sen1)==0)
{
    printf("palindrome\n");
}
}
int main()
{
    int ch;
    printf("choice 1 is to find the length of a string(not using a in built function)\n");
    printf("choice 2 is to print each character of a string in a new line\n");
    printf("choice 3 is to Count vowels and consonants in a string.\n");
    printf("exit is choice 4\n");
    printf("enter your choice\n");
    scanf("%d",&ch);
    while (getchar() != '\n');
    switch (ch)
    {
        case 1:

```

```
{  
    length();  
    break;  
}  
case 2:  
{  
    print();  
    break;  
}  
case 3:  
{  
    vowcon();  
    return 0;  
}  
case 4:  
{  
    printf("exit\n");  
    return 0;  
}  
default:  
{  
    printf("invalid choice\n");  
    break;  
}  
}  
return 0;  
}
```